*Tectonophysics*, 104 (1984) 1–34
Elsevier Science Publishers B.V., Amsterdam – Printed in The Netherlands



# PALAEOMAGNETISM OF CALEDONIAN INTRUSIVE SUITES IN THE NORTHERN HIGHLANDS OF SCOTLAND: CONSTRAINTS TO TECTONIC MOVEMENTS WITHIN THE CALEDONIAN OROGENIC BELT

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(Received March 23, 1983; revised version accepted September 6, 1983)

## ABSTRACT

Esang, C.B. and Piper, J.D.A., 1984. Palaeomagnetism of Caledonian intrusive suites in the Northern Highlands of Scotland: constraints to tectonic movements within the Caledonian orogenic belt. *Tectonophysics*, 104: 1–34.

To evaluate the scale of tectonic movements within the northern sector of the 500–400 Ma Caledonian orogenic belt and its Precambrian foreland zone between the Great Glen Fault (GGF) zone to the southeast and the Laurentian Block to the northwest, we have studied the palaeomagnetism of minor intrusive rocks within the Northern Highlands terrain. These rocks include (1) amphibolites and other metamorphic rocks predating deformation, (2) microdiorities, dolerites and related suites emplaced, and probably magnetised, between 450 and 420 Ma, and (3) Lower–Middle Devonian lamprophyres. A range of predominantly NNE negative and SSW positive components are resolved by cleaning treatment with a dispersion of declinations towards a minority WNW–ESE axis; isolated southerly negative directed hematite-held components suggests limited, but no widespread, remagnetisation in Devonian–Carboniferous times.

Comparison is made with data from other tectonic divisions in the Caledonian orogenic belt and the bordering forelands. Palaeopoles from the Northern Highlands closely conform in part with North American Ordovician poles and in part with the post-Ordovician palaeopoles from Britain south of the GGF. The definitive motions of the British Caledonides to emerge from the palaeomagnetic analysis are an anticlockwise rotation of the Caledonian terrain in early Ordovician times, small relative motions during the remainder of Ordovician times followed by large clockwise and then anticlockwise rotations during late Ordovician to early Silurian times contemporary with the last major movements on the Moine Thrust (ca. 430 Ma). Late Silurian–Devonian movements along the GGF were probably below the limits of palaeomagnetic detectability. The collective data require that apparent polar wander movements and concomitant continental movements have currently been incompletely recovered by North American studies and the path for Lower Palaeozoic times is more complex than recognised hitherto.

## INTRODUCTION

Formation of the Caledonian Orogenic belt was genetically linked to the consumption of a major ocean basin (Iapetus) in Ordovician and Silurian times. The

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closure and demise of this ocean is evidenced by the faunal differences between the bordering continents (e.g. Cowie, 1971; Cocks and Fortey, 1982), the obduction of ophiolite slices of primary Cambrian and Ordovician age (Dewey, 1974) and magmatism linked to a subducting ocean plate (Fitton and Hughes, 1970). It is, however, paradoxical that palaeomagnetic studies have currently only succeeded in demonstrating modest separations between the bordering margins of the Iapetus Ocean as defined by the ophiolite occurrences, while more important apparent differences are identified between tectonic divisions separated by no obvious major ($>$ 1000 km of movement) tectonic boundaries (Briden et al., 1973; Morris, 1976; Piper, 1979; Deutsch, 1982). As noted by Yardley et al. (1982), models reconstructing plate tectonic evolution use present-day geological elements, a conservative approach which considers plate motions essentially in terms of back and forth motions. The lack of scope for strike-slip movements means that these models are unlikely to be correct. Strike-slip tectonics are known for example to be critical to the geological evolution of many circum-Pacific environments (e.g., Beck 1980). They are also known to be relevant to the Caledonides (Pitcher 1969) but the geological constraints here suggest movements below the limits of palaeomagnetic detection.

The scope for involving large post-Lower Ordovician closure between Caledonian tectonic components in the British Isles is restricted because the palaeomagnetic record from the orthotectonic Grampian Highlands is closely comparable to the contemporary record from the paratectonic Caledonides of England and Wales (Briden et al., 1973), and is compatible with a separation across Iapetus close to the statistical limits of the data. The collective data are, however, rather insensitive to any possible strike-slip motions because the magnetic vectors are mostly at high angles to the NE–SW to ENE–WSW trend of the Caledonian tectonic boundaries. It is when the British data are compared with the contemporary North American and European palaeomagnetic evidence on the post-Caledonian continental reconstruction (Bullard et al., 1965) that severe problems arise: an apparent discrepancy equivalent to some 20° of palaeolatitude (Briden et al., 1973) persists from early Ordovician to late Lower Devonian times (Piper, 1979). The tectonic equivalent is a sinistral strike-slip motion of some 2000–3000 km (Morris, 1976; Piper, 1979).

There is some geological evidence for appreciable strike-slip motion across the major Caledonian boundaries north of the Iapetus line. For example, Moinian sediments in the Northern Highlands lie unconformably on Lewisian basement and were influenced by Proterozoic magmatism (ca. 1050 Ma) and folding closely contemporary with the Grenville mobile episode in the Laurentian Shield. South of the Great Glen Fault, Moinian sediments continue, apparently without major unconformity, into the Dalradian succession which includes early Palaeozoic sediments (Harris and Pitcher, 1975). The Great Glen is the largest of a set of NE–SW strike-slip faults with an important component of early Devonian movement (Pitcher, 1969), but the other members of this set do not appear to displace Moinian–Dalradian

features in a major way and are therefore unlikely to have produced movements capable of palaeomagnetic detection. Further south there is no presently-adjacent source for the southerly-derived Southern Highland Group sediments in the Grampians (Yardley et al., 1982), but no palaeomagnetic case has yet been made for major strike-slip motion along the Highland boundary zone because data from the Grampians, specifically the Younger Gabbros of Aberdeenshire, conform closely to those from the remainder of Britain (Briden et al., 1973). Thus the Great Glen Fault is the only apparent contender for accommodating a large amount of strike-slip motion between the British sub-plate on one hand and North America on the other. To be successfully integrated with the geological constraints it must be demonstrated that any detectable strike-slip motion is pre-Middle Devonian in age, because the continuity of Middle and Upper Old Red Sandstone sedimentation across the line of the Great Glen Fault (GGF) precludes major movement after this time (Mykura, 1975; Donovan et al., 1976).

However, recently Scotese and Van der Voo (1982) have dismissed these geological constraints to argue on palaeomagnetic grounds alone for 2000 km of intra-Carboniferous sinistral strike-slip motion on the fault zone. Their analysis is more embracing than an earlier case put by Storetvedt (1975) who considered a few selected palaeopoles interpreted in the context of a strike slip motion which is less than the effective resolution of the palaeomagnetic technique (Turner et al., 1976). Van der Voo and Scotese argue that magnetisations derived from Old Red Sandstone sediments and igneous rocks north of the GGF are primary and show that they are included within a group of poles distinct from contemporary data south of the GGF. In part this case has been prompted by an attempt to integrate apparent movements between the Acadian Province of the Appalachian terrain (incorporating the maritime provinces of the U.S.A. and Canada) and the rest of North America, where a palaeomagnetic discrepancy may be present until mid-Carboniferous times (Kent and Opdyke, 1978). The real or apparent significance of this difference has been subject to contentious debate (see Roy, 1982, and Kent and Opdyke, 1982, for recent discussions) but it is nevertheless curious that the model derived, namely 2000–3000 km of sinistral strike-slip movement, is statistically-identical to the model derived from the pre-Middle Devonian data from Britain and North America (Morris, 1976; Piper, 1979); only the timing is different, one requiring a Lower–Middle Devonian motion (which does not violate the Middle–Upper Old Red Sandstone geological constraints, but cannot be well tested against older constraints) and the other requiring a mid-Carboniferous motion which clearly does violate the geological constraints.

This conflict can only be properly evaluated by improving the palaeomagnetic data base and more particularly by establishing the pre-Middle Devonian apparent polar wander (a.p.w.) record from the Caledonides north of the GGF. With this objective in mind we have sampled a range of Caledonian rocks from the Northern Highlands and analysed their magnetisations. These rocks are incorporated within

terrain deformed by a number of episodes up to Silurian times (ca. 430–410 Ma) and uplifted, cooled and closed to the diffusion of argon between 480 and 400 Ma (Dewey and Pankhurst, 1970).

## GEOLOGICAL OUTLINE

### *Regional setting*

In the Northern Highlands the Moinian country rocks comprise predominantly arenaceous quartz-feldspathic and mica schists, derived from sediments deposited on a subsiding shelf (Richey and Kennedy, 1939; Phemister, 1961; Bennison and Wright, 1969; conglomeratic facies overlying Lewisian-type rocks in inliers suggest a broad continuity of the basement with the Precambrian foreland northwest of the Moine Thrust Belt.

Although the Moinian sediments were profoundly deformed by Caledonian events there is a growing body of evidence that they were effected by Upper Proterozoic events as well. "Morarian" ages of 780–740 Ma have been established from pegmetites of Knoydart and Moine rocks of the Morar Division (Long and Lambert, 1963; Bowes, 1968; Lambert, 1969; Van Breemen et al., 1974 and Johnstone, 1975), and Rb–Sr isochron ages of 720 ± 120 Ma and 700 ± 100 Ma have been obtained from semi-pelites in the Morar Division by van Breemen et al. (1978). An earlier episode is recognised by Brook et al. (1977) and Brewer et al. (1979): the Ardgour granite gneiss, a migmatite believed to have formed in situ from surrounding migmatitic metasediments yields a Rb–Sr age of 1010 ± 50 Ma (Brook et al., 1976) and the southwestern Moines yield a domain isochron of 1004 ± 28 Ma (Brewer et al., 1979); granitic gneiss at Loch Quoich yields an age of ca. 1000 Ma (Van Breemen and Piasecki, 1980). Palaeomagnetic work has subsequently established that the Lower and Upper Torridonian sediments are ca. 1100 and 1040 Ma in age respectively (Smith et al., 1982) and restores the classical correlation of the Moinian and Torridonian sediments (e.g., Giekie, 1892). In a wider context these sediments are contemporary in time and place with initial break up of the Laurentian and Fennoscandian Shields immediately post-dating the Grenville episode at ca. 1100 Ma, an event which integrates together the subsidence of the Lewisian shelf, the intra-Torridonian deformation and marine incursion at the top of the Torridonian, and the ca. 1050–1000 Ma magmatism in the Moines (Piper 1983).

During subsequent Caledonian deformation highest bulk strains occurred during the first phase of deformation characterised by large recumbent nappes; a consequence of this deformation is the Moine Thrust Zone. A second phase refolded the nappes to produce steeply inclined axial planes dipping eastwards. A third phase produced large regional antiforms which change progressively in character towards the central part of the Moine outcrop. The development of the nappes resulted in metamorphism at low temperatures (400°–200°C) and a moderate load pressure

(2–4 kbar) (e.g., Johnson and Shepherd, 1970; Dewey and Pankhurst, 1970), but during the late stages of tectonism high temperatures (600°–500°C) led to recrystallisation (andalusite to kyanite together with stauralite) characterising the Buchan-type metamorphic assemblage and producing the migmatite core of the orogen trending NNE. A maximum temperature of 680°C was achieved at this stage (Richardson, 1968). Kyanite indicates a depth of metamorphism in excess of 12 km but not exceeding 20 km due to stacking of nappe structures (Zwart, 1967). The third structural event is associated with the beginning of retrogressive metamorphism.

The subsequent phase of post-tectonic uplift and cooling is documented by isotopic closure as mineral systems closed to migration of argon. "Thermochron" ages for K–Ar mineral ages from the Moine Schists increase progressively towards the Moine Thrust Zone from about 410 to 562 Ma (Fig. 1) in a pattern reckoned to reflect regional cooling rather than a variable overprint of older systems (Dewey and Pankhurst 1970). Muscovite, which becomes closed to argon at 300°–200°C, records K–Ar ages averaging about 10 Ma older than biotite which becomes closed at 250°–150°C (Harper 1967, Dewey and Pankhurst 1970); assuming a geothermal gradient of 30°–35°C $km^{-1}$ muscovite would have become closed at 7–4 km depth and biotite at 5–3 km depth, and the uplift magnetisations were therefore acquired at greater depths than these. Dewey and Pankhurst (1970) accept the maximum rate of subsidence of 0.25 km/Ma in the South Mayo Trough as a possible maximum uplift rate in the Scottish Highlands, close to the lower end of the range of uplift rates of 0.8–0.2 km/Ma from Alpine and Andean orogenic belts.

*The Caledonian intrusive suites*

Minor intrusions, in the form of narrow parallel-sided bodies ranging from dykes to low-angle sheets, are widely distributed in the Northern Highlands (Smith, 1979) being very numerous in the southwestern sector and less important towards the north-northeast. They are subdivided into a number of suites by intersecting relationships each one representing a phase of magmatic injection into older metamorphic rocks. The earliest suite intruded between the first and second deformation trends sub-parallel to the fold belt (Smith 1979); it was metamorphosed along with the country rocks and is now represented by amphibolite bodies. A second group postdates intrusion of major complexes and is numerically the most important; it ranges from ultrabasic types to microdiorites (very numerous and comprising easterly-dipping sheets) to appinites (restricted) and to acid porphyrites in composition. These intrusions frequently exhibit internal foliation or lineation oblique to the margins due to shearing during consolidation late in the Caledonian tectonism; some are thoroughly recrystallised and the suite appears to have been largely or wholly intruded while the metamorphic terrain was still at high temperatures. This suite was in turn succeeded by minnette lamprophyres and felsites as typical igneous dykes and sheets showing chilled margins and little or no deformation. They have generally

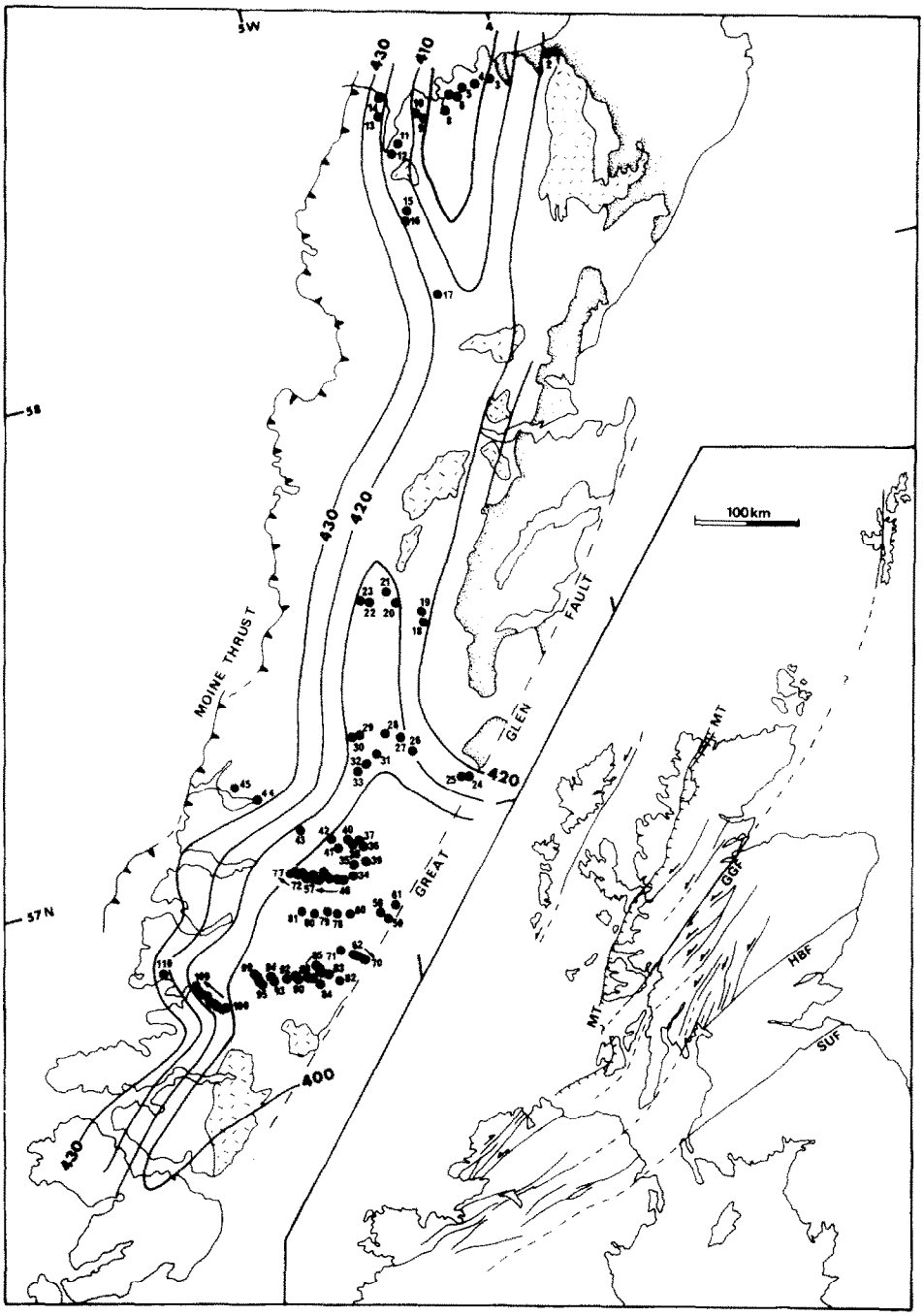



Fig. 1. Location of palaeomagnetic sampling sites in the Caledonian terrain of Northwest Scotland. The chrontour zones are taken from Dewey and Pankhurst (1970) and are based on muscovite K–Ar mineral closure dates. The inset diagram shows the system of major late Caledonian (Late Silurian–Devonian) age faults cutting the orthotectonic Caledonides after Pitcher (1969). *MT* = Moine Thrust, *GGF* = Great Glen Fault, *HBF* = Highland Boundary Fault, *SUF* = Southern Uplands Fault.

been considered to be of Lower Old Red Sandstone age and genetically linked to the younger granites; they cut most of these granites and occur as pebbles in the basal conglomerates of the Middle Old Red Sandstone sediments. The amphibolite suite was intruded between the Carn Chuinneag granite (ca. 560 Ma) and the late Caledonian pegmatites at ca. 450 Ma (Smith, 1979). The microdiorite–appinite suite cuts the 450 Ma pegmatites and predates the Strontian and Ross of Mull granites intruded at 420 Ma (Beckinsale and Obradovich, 1973; Smith, 1979).

Thus the rocks of this study (excepting the lamprophyres) were uplifted and cooled with the regional metamorphic terrain and have a magnetic remanence record related, at least in part, to this post-tectonic late Caledonian event. Theoretical aspects of the acquisition of remanence during slow cooling have been treated by a number of workers (Ueno et al., 1975; Pullaiah et al., 1975; Dodson and McClelland Brown, 1980). The analysis of the latter workers shows that the magnetic blocking temperatures of rocks subject to the cooling rates of the Caledonides are only slightly lower than the unblocking temperatures determined in the laboratory implying that the uplift magnetisations are appreciably older than the K–Ar mineral ages. Some practical studies of uplift magnetisations in Precambrian rocks (Morgan 1976, Piper 1981) identify some of the main characteristics of remanence acquisition under conditions of slow cooling, namely the averaging of short term secular features of the magnetic field and the recording of time sequences of palaeopoles by higher to lower blocking temperature phases and earlier to later-cooled facies. Uplift rates in the Caledonides were at least an order more rapid than these Proterozoic terrains (Piper, 1981; Dewey and Pankhurst, 1970) and two features of palaeomagnetic behaviour characterise this Caledonian collection; the first is a tendancy for single units to record directions of both normal and reversed polarity generally in different phases (magnetite and hematite) but sometimes in the same phase, and the second is a tendancy for sample directions within individual sites to be smeared in a distribution which is not effectively reduced by cleaning and frequently lies along part of a great circle incorporating normal and reversed field axes. Systematic migration of sample directions back along an ancient a.p.w. trend with progressive cleaning (Morgan, 1976) has not been positively identified possibly because the cooling rate was too fast for blocking temperature spectra to record a significant path length.

## PALAEOMAGNETIC RESULTS

### *Field and laboratory methods*

Cores drilled in the field were oriented by sights on the Sun and/or magnetic bearings in conjunction with topographic sights. A site normally comprises six or seven independently-oriented cores from one geological unit but up to 30 cores were drilled at some sites in Moinian metasediments. In the laboratory the alternating field (a.f.) method was employed as the routine method of treatment because it tends

TABLE I

Palaeomagnetic results from Caledonian intrusive suites, Northern Highlands of Scotland *

| Site No. | Rock division | Grid ref | Chrontour zone | | $D$ | $I$ | $N/n/R$ | Character of component |
|---|---|---|---|---|---|---|---|---|
| 1 | MDA | 2801 9605 | 2 | | 8 | −34 | 9/8/7.69 | 3 2 * M |
| 2 | MDA | 2906 9605 | 2 | | 6 | −12 | 6/6/5.78 | 2 1 M+H |
| 3 | MDA | 2709 9604 | 4 | | 173 | 7 | 7/6/5.90 | 2 1 * M |
| 4 | ALI | 2706 9604 | 5 | | 325 | −11 | 10/10/9.81 | 1 1 H |
| 5 | ALI | 2703 9603 | 5 | | 282 | 32 | 7/6/5.42 | 1 3 M |
| 6 | MDA | 2906 9604 | 5 | (i) | 285 | −36 | 7/2/1.95 | 3 2 M |
| | | | | (ii) | 10 | −34 | 7/3/2.93 | 3 3 – |
| 7 | M | 2701 9507 | 5 | | 292 | −32 | 8/3/2.81 | 3 3 – |
| 8 | ALI | 2606 9508 | 5 | | 12 | −21 | 6/6/5.94 | 1 1 M |
| 8a | MDA | 2606 9508 | 5 | | 3 | −34 | 6/6/5.84 | 2 1 M |
| 9 | ALI | 2604 9509 | 4 | | 183 | 16 | 7/4/3.84 | 3 3 M |
| 10 | D(LI) | 2508 9600 | 4 | | 171 | 27 | 7/7/6.95 | 2 1 H |
| 12 | D | 2509 9305 | 3 | (i) | 29 | 32 | 8/6/5.10 | 3 2 M |
| | | | | (ii) | 305 | −28 | 8/3/2.85 | – 2 H |
| 14 | ALI | 2602 9306 | 3 | | 21 | −37 | 11/4/3.61 | 1 2 * – |
| 15 | MDA | 2601 9305 | 3 | | 196 | 46 | 7/5/4.79 | |
| 16 | MDA | 2702 9507 | 3 | (i) | 18 | 34 | 7/4/3.70 | 3 2 – |
| | | | | (ii) | 31 | −23 | 7/2/1.99 | 3 1 H |
| 17 | MDA | 2705 9602 | 3 | | 174 | 3 | 7/5/4.4 | 3 1 M |
| 18 | D | 2405 8506 | 3 | | 353 | −35 | 10/3/2.89 | 2 3 M |
| 19 | MDA | 2405 8507 | 3 | | 175 | −7 | 9/9/8.89 | 2 1 M |
| 20 | M | 2308 8602 | 4 | | 123 | 29 | 7/6/5.19 | 3 3 – |
| 21 | M | 2305 8603 | 4 | (i) | 275 | −6 | 10/7/6.57 | 3 3 M |
| | | | | (ii) | 193 | 24 | 10/2/1.98 | –2 H |
| 22 | M | 2301 8601 | 4 | | 283 | −18 | 17/17/15.40 | 3 2 – |
| 23 | M | 2209 8602 | 4 | | 49 | 37 | 13/8/7.68 | 3 2 – |
| 24 | S | 2501 8209 | 3 | | 186 | −43 | 7/7/6.72 | 3 1 h |
| 25 | MDA | 2408 8300 | 3 | | 211 | 26 | 6/4/3.75 | 2 1 M |
| 26 | MDA | 2307 8301 | 3 | | 190 | 17 | 7/5/4.85 | 3 3 M |
| 27 | M | 2305 8302 | 4 | (i) | 101 | 34 | 10/5/4.83 | 3 1 – |
| | | | | (ii) | 171 | 0 | 10/4/3.88 | 3 1 – |
| 28 | M | 2300 8302 | 4 | (i) | 296 | 9 | 15/12/9.23 | 3 2 M |
| | | | | (ii) | 191 | 4 | 15/5/4.81 | – 2 H |
| 29 | M | 2204 8302 | 4 | (i) | 103 | 1 | 13/8/6.93 | 3 2 – |
| | | | | (ii) | 170 | 21 | –/2/1.99 | 3 3 – |
| 30 | M | 2202 8301 | 4 | (i) | 299 | −35 | 14/6/5.76 | 3 2 M |
| | | | | (ii) | 24 | −17 | 14/8/7.63 | 3 2 – |
| | | | | (iii) | 15 | 2 | 14/3/2.95 | 3 2 – |
| 31 | M | 2208 8208 | 4 | (i) | 10 | −34 | 13/9/8.77 | 2 2 M |
| | | | | (ii) | 303 | −22 | 13/6/5.1 | – 2 H |

TABLE I (continued)

| Site No. | Rock division | Grid ref | Chrontour zone | | *D* | *I* | *N/n/R* | Character of component |
|---|---|---|---|---|---|---|---|---|
| 32 | M | 2205 8206 | 4 | | 126 | 29 | 8/7/5.82 | 3 2 * – |
| 33 | M | 2204 8205 | 4 | (i) | 166 | 27 | 10/8/7.78 | 2 1 M |
| | | | | (ii) | 114 | 2 | 10/2/1.85 | 2 3 – |
| 34 | MDA | 2265 8207 | 5 | | 25 | −37 | 6/5/4.97 | 2 1 M |
| 35 | MDA | 2200 8095 | 5 | | 182 | 21 | 8/7/6.63 | 3 1 M |
| 36 | D | 2109 8099 | 5 | | 20 | −35 | 9/7/6.79 | 2 1 M |
| 38 | D | 2108 8100 | 5 | | 333 | −28 | 6/2/1.95 | 3 3 M |
| 39 | D | 2202 8097 | 5 | | 5 | −32 | 8/5/4.62 | 3 2 * M |
| 40 | MDA | 2108 8101 | 5 | | 16 | −37 | 7/7/6.38 | 3 2 M |
| 41 | MDA | 2105 8099 | 5 | | 0 | −48 | 7/7/6.88 | 3 1 M |
| 42 | MDA | 2101 8101 | 4 | | 0 | −44 | 7/6/5.93 | 3 2 M |
| 43 | MDA | 2002 8102 | 4 | (i) | 10 | −42 | 7/5/4.72 | 1 2 * M |
| | | | | (ii) | 5 | 7 | −/−/1 | – 1 H |
| 44 | G | 1905 8200 | 1 | | 316 | −36 | 7/6/5.70 | 2 1 H |
| 45 | MDA | 1901 8203 | 1 | | 214 | 31 | 6/6/5.73 | 3 1 Mh |
| 46 | MDA | 2257 8027 | 5 | | 145 | 15 | 6/5/4.72 | 2 2 * M |
| 46a | MDA | 2265 8027 | 5 | | 25 | −37 | 6/5/4.97 | 2 1 M |
| 47 | MDA | 2256 8027 | 5 | | 21 | −19 | 6/4/3.72 | 3 1 * M |
| 48 | MDA | 2224 8024 | 5 | | 8 | −5 | 8/5/4.85 | 2 2 M |
| 49 | MDA | 2224 8029 | 5 | | 331 | 35 | 6/3/2.97 | 3 2 M |
| 50 | MDA | 2205 8035 | 5 | | 358 | −57 | 6/5/4.12 | 1 2 M |
| 51 | MDA | 2139 8013 | 5 | | 295 | −5 | 7/4/3.60 | 2 2 M |
| 52 | MDA | 2132 8017 | 5 | | 183 | 19 | 7/2/1.99 | 3 1 – |
| 53 | MDA | 2135 8015 | 5 | (i) | 105 | 62 | 6/3/2.99 | 2 2 M |
| | | | | (ii) | 17 | −4 | 6/4/3.81 | 2 2 – |
| 54 | MDA | 2132 8017 | 5 | (i) | 284 | 24 | 7/3/2.93 | 2 1 M |
| | | | | (ii) | 23 | −24 | 7/4/3.99 | – 1 M |
| 55 | MDA | 2118 8015 | 5 | | 34 | −59 | 7/6/5.90 | 2 1 M |
| 56 | MDA | 2112 8019 | 5 | | 286 | −34 | 7/5/4.76 | 1 1 M |
| 57 | D | 2085 8019 | 5 | | 316 | −17 | 7/4/3.68 | 3 2 M |
| 57a | MDA | 2092 8022 | 5 | | 192 | 18 | 8/3/2.95 | 3 1 M |
| 58 | MDA | 2175 7887 | 5 | | 316 | −16 | 7/7/6.02 | 1 2 M |
| 58a | MDA | 2176 7890 | 5 | | 175 | 48 | 5/5/4.85 | 1 2 M |
| 58b | L | 2177 7891 | 5 | (i) | 197 | 38 | 6/2/1.97 | 3 2 M |
| | | | | (ii) | 321 | −42 | 6/4/3.85 | 2 1 M |
| 59 | MDA | 2187 7868 | 5 | | 5 | −42 | 5/4/3.85 | 1 1 M |
| 59a | D | 2198 7889 | 5 | | 33 | 43 | 6/6/5.50 | 3 2 – |
| 60 | MDA | 2126 7910 | 5 | | 316 | −48 | 9/9/8.50 | 2 1 M |
| 61 | D | 2198 7889 | 5 | | 174 | 40 | 10/10/9.92 | 1 1 M |
| 62 | MDA | 2118 7834 | 5 | | 127 | 56 | 7/5/4.87 | 1 1 M |
| 63 | MDA | 2122 7833 | 5 | | 26 | 10 | 7/7/6.38 | 1 1 M |

TABLE I (continued)

| Site No. | Rock division | Grid ref | Chrontour zone | | *D* | *I* | *N*/*n*/*R* | Character of component |
|---|---|---|---|---|---|---|---|---|
| 64 | D | 2123 7832 | 5 | (i) | 246 | 39 | 5/5/4.76 | 1 1 M |
| | | | | (ii) | 194 | −39 | −/−/1 | − 1 H |
| 65 | MDA | 2601 9305 | 5 | | 196 | 46 | 7/5/4.79 | 2 1 M |
| 66 | DGL | 2136 7827 | 5 | | 189 | −14 | 6/7/6.92 | 1 1 M |
| 67 | D | 2139 7825 | 5 | (i) | 190 | −7 | 5/5/4.61 | 1 1 M |
| | | | | (ii) | 319 | −54 | 5/3/2.93 | 1 2 M |
| 68 | MDA | 2143 7821 | 5 | (i) | 253 | 8 | 6/6/5.85 | 1 1 M |
| | | | | (ii) | 260 | 16 | 6/3/2.99 | − 2 H |
| 69 | DGL | 2137 7825 | 5 | | 301 | −37 | 7/6/5.67 | 1 2 M |
| 70 | MDA | 2146 7819 | 5 | | 347 | −38 | 6/6/5.96 | 1 1 Mh |
| 71 | D | 2095 7848 | 5 | | 211 | 43 | 7/5/4.87 | 1 1 M |
| 72 | MDA | 2072 8025 | 5 | | 212 | 29 | 7/6/5.92 | 2 1 * M |
| 73 | D | 2069 8025 | 5 | | 56 | −50 | 6/5/4.88 | 3 1 M |
| 74 | M | 2029 8030 | 5 | (i) | 202 | 10 | 12/7/6.69 | 2 1 M |
| | | | | (ii) | 325 | 3 | 12/2/1.96 | 2 2 M |
| 75 | D | 2017 8040 | 4 | (i) | 13 | 23 | 7/5/4.60 | 1 2 − |
| | | | | (ii) | 100 | 34 | 7/3/2.97 | 1 3 p M |
| 76 | M | 2014 8040 | 4 | (i) | 94 | 35 | 7/3/2.86 | 3 2 − |
| | | | | (ii) | 7 | −36 | 7/3/2.93 | 3 1 M |
| 77 | MDA | 2004 8033 | 4 | | 347 | −40 | 7/3/2.96 | 2 1 M |
| 78a | MDA | 2060 7922 | 5 | | 307 | −35 | 8/6/5.50 | 3 2 − |
| 78b | D | 2053 7923 | 5 | | 29 | −49 | 6/5/4.86 | 3 1 M |
| 79 | D | 2043 7924 | 5 | | 23 | −27 | 6/3/2.89 | 3 2 M |
| 80 | D | 2019 7921 | 5 | | 28 | −22 | 6/5/4.62 | 2 2 M |
| 80a | MDA | 2018 7921 | 5 | | 134 | 37 | 7/7/6.73 | 1 1 M |
| 81 | D | 2006 7927 | 5 | | 34 | −51 | 7/7/6.99 | 3 1 M |
| 82 | D | 2052 7784 | 5 | | 16 | −46 | 7/6/5.92 | 3 3 M |
| 83 | D | 2022 7701 | 5 | | 340 | −29 | 8/7/6.99 | 3 1 M |
| 84 | D | 2021 7789 | 5 | | 345 | −24 | 8/8/7.92 | 2 1 M |
| 85 | D | 2022 7795 | 5 | | 349 | −33 | 6/6/5.98 | 2 1 M |
| 86 | D | 2022 7708 | 5 | | 286 | −34 | 8/7/6.85 | 1 2 M |
| 87 | D | 2021 7792 | 5 | | 348 | −28 | 7/7/6.79 | 2 1 M |
| 88 | D | 2022 7807 | 5 | | 192 | −12 | 6/2/1.98 | 2 3 M |
| 89 | MDA | 2019 7811 | 5 | | 0 | −38 | 8/8/7.61 | 1 1 M |
| 90 | MDA | 1988 7788 | 5 | | 202 | 23 | 8/8/7.63 | 2 2 * M |
| 91 | D | 1976 7792 | 5 | | 124 | 13 | 6/5/4.90 | 1 2 M |
| 92 | L | 1974 7793 | 5 | | 183 | −45 | 6/6/5.96 | 2 1 M |
| 93 | MDA | 1947 7794 | 5 | | 358 | −54 | 6/7/6.96 | 2 1 M |
| 94 | MDA + M | 1927 7797 | 5 | | 17 | −50 | 7/7/6.84 | 2 1 M, h |
| 94a | D | 1933 7796 | 5 | | 334 | −39 | 5/3/2.94 | 3 2 M |
| 95 | D | 1974 7793 | 5 | | 11 | 2 | 6/5/4.66 | 3 3 M |
| 96 | MDA | 1920 7801 | 5 | | 274 | −13 | 6/6/5.11 | 3 2 * M |

TABLE I (continued)

| Site No. | Rock division | Grid ref | Chrontour zone | | $D$ | $I$ | $N/n/R$ | Character of component |
|---|---|---|---|---|---|---|---|---|
| 97 | D | 1915 7803 | 5 | (i) | 16 | −46 | 6/6/5.92 | 2 1 M |
| | | | | (ii) | 190 | −42 | 6/5/4.64 | - 2 H |
| 98 | D | 1914 7803 | 5 | | 8 | −43 | 6/6/5.87 | 2 1 p, M |
| 99 | MDA | 1908 7807 | 5 | | 22 | −48 | 6/6/5.98 | 1 1 M |
| 100 | L | 1869 7816 | 4 | | 192 | −38 | 6/5/4.96 | 2 1 M+H |
| 101 | L | 1868 7816 | 4 | | 300 | −47 | 6/5/4.59 | 2 1 M |
| 102 | L | 1855 7815 | 4 | | 188 | −16 | 6/5/4.97 | 1 1 M |
| 103 | L | 1855 7816 | 4 | | 5 | 9 | 6/5/4.87 | 2 2 * M |
| 104 | D | 1844 7815 | 4 | | 279 | 15 | 8/6/5.05 | 2 1 M |
| 105 | L | 1821 7819 | 4 | | 132 | 27 | 7/7/6.68 | 1 2 * M |
| 106 | L | 1808 7827 | 4 | | 25 | 0 | 8/8/7.76 | 2 1 M |
| 107 | L | 1789 7828 | 4 | | 125 | 18 | 8/3/2.89 | 3 3 M |
| 108 | L | 1773 7829 | 4 | | 199 | −17 | 7/4/3.99 | 1 1 Mh |
| 109 | MDA | 1767 7824 | 4 | | 22 | −17 | 7/7/6.99 | 1 1 Mh |
| 110 | MDA | 1717 7844 | 2 | (i) | 9 | −20 | 8/13/12.81 | 1 1 M |
| | | | | (ii) | 188 | −38 | -/-/1 | 1 1 H |

* Rock type subdivisions are abbreviated: D = dolerites as sheets or dykes; MDA = member of diorite, microdiorite and appinite suite of minor intrusions (mostly dykes and sheets); ALI = amphibolite facies basic divisions in Lewisian inlier; M = Moinian metasediments; S = serpentinite intrusion; G = microgranite; DGL = major diorite complex at Glen Loy; L = Lamprophyre. The regional muscovite and whole rock chrontour zones are numbered: (1) > 430 Ma; (2) 430–420 Ma; (3) 420–410 Ma; (4) 410–400 Ma; (5) < 400 Ma. $D$ and $I$ are the mean declination and inclination respectively of $n$ samples illustrating the component out of a total of $N$ samples from the site. The character of the component is described in the last column in terms of two numbers and a letter. For the first number: 1 = strong Total NRM, > 50 A $m^2$ $kg^{-1}$, 2 = intermediate intensity Total NRM, 50–10 A $m^2$ $kg^{-1}$, 3 = weak Total NRM, < 10 A $m^2$ $kg^{-1}$. For the second number: 1 = highly stable to progressive treatment with > 90% of steps lying along single straight lines of vector plots; these samples usually intermediate to high coercivities. 2 = intermediate stability to progressive treatment with 50–90% of steps lying along the vector plots. 3 = low stability with < 50% of steps lying along straight lines of vector plots defining the mean direction; when these yield a precision parameter less than 10 they are rejected. M = Magnetite Curie point defined by a.f. cleaning behaviour plus thermomagnetic study and/or second stage thermal treatment; p = probable pyrrhotite Curie point at ca. 320°C; H = hematite component identified by thermal cleaning above the Curie point of magnetite; h = hematite content indicated by thermomagnetic analysis but not positively linked to any particular component.

to be most discriminating for the broad ranges of coercivities usually present in igneous and metamorphic rocks. Cleaning fields were incremented in steps of 5 or 10 milliTesla (mT) up to maximum peak fields of 140 mT. Thermal demagnetisation was applied to a pilot sample from each site in steps of 150°C to 450°C and 50°C reduced to 20°C above this to the Curie points of magnetite (580°C) or hematite

(680°C); this was frequently followed by the application of thermal cleaning to the remainder of the a.f. cleaned collection from each site at four or five steps above 450°C. Stable end points were resolved from the successive linear segments of the vector plots. Site group means are listed in Table I and an outline of the salient features of the demagnetisation results follows below.

*Diorites and related intrusions*

Whilst the material from this and the other rock groups shows collectively a good degree of coherent directional behaviour with treatment, the magnetic record is complex both at between-site and within-site levels. Frequently only a proportion of samples illustrate stable behaviour while at other sites individual stability is not reflected in significant between-sample groupings. For this reason the calculated directions (Table I) are often based on only part of the site collection. We group sites in the diorites and related rocks according to their behaviour against stepwise a.f. and thermal treatment into five divisions, namely: (i) sites which have weak NRM's of poor stability (1, 6, 26 and 96); (ii) sites with weak but stable NRM's (16, 17, 25, 40, 45, 46, 48, 49, 57A, 60, 72 and 90); (iii) sites with weak and stable NRM's showing high sample coercivity and within-sample grouping (2, 3, 41, 42, 43, 47, 55 and 77) which is not always reflected in between-sample grouping; (iv) sites with strong low coercivity NRM's of variable stability (50, 53, 56, 58, 62, 63, 89, 94, 99 and 101); and (v) sites with strong high coercivity and stable NRM's (19, 34, 44, 54, 59, 66, 68, 69, 70, 93 and 110).

Sites in the first group have only some cores sufficiently strongly magnetised to be reliably measured ($> 2 \cdot 10^{-6}$ A m$^2$ kg$^{-1}$) and NRM's are generally reduced below this range by treatment in fields of 20–30 mT, normally following removal of a low coercivity component. Samples in the second group have a sufficiently broad range of coercivities and blocking temperatures to permit more detailed analysis. Site 72 is a noteworthy example because three of six cores reverse polarity during cleaning (Fig. 2), changing from northerly shallow to a southerly positive direction unblocked at 560°C in the thermal pilot; primary cooling through the critical range of blocking temperatures may have recorded a reversal here. An example of weak but stable behaviour following removal of a low coercivity component is illustrated at site 54 (Fig. 2). A common response to cleaning is movement from a steep northerly positive to less steep northerly positive/negative direction (e.g. 48) but contrasting end points are yielded at sites such as 60 where stable end points are north westerly negative, a direction recognised in smaller numbers of cores from several other sites (e.g. 60, Fig. 2) and stable to thermal treatment. At site 16, (i) is a low coercivity component, possibly of composite origin, removed at low temperatures, and (ii) is defined by two samples above the Curie point of magnetite. The third group of sites exhibit systematic behaviour residing in medium to high coercivity components. At sites 41 (Fig. 2) and 43 all samples move from northerly positive to northerly

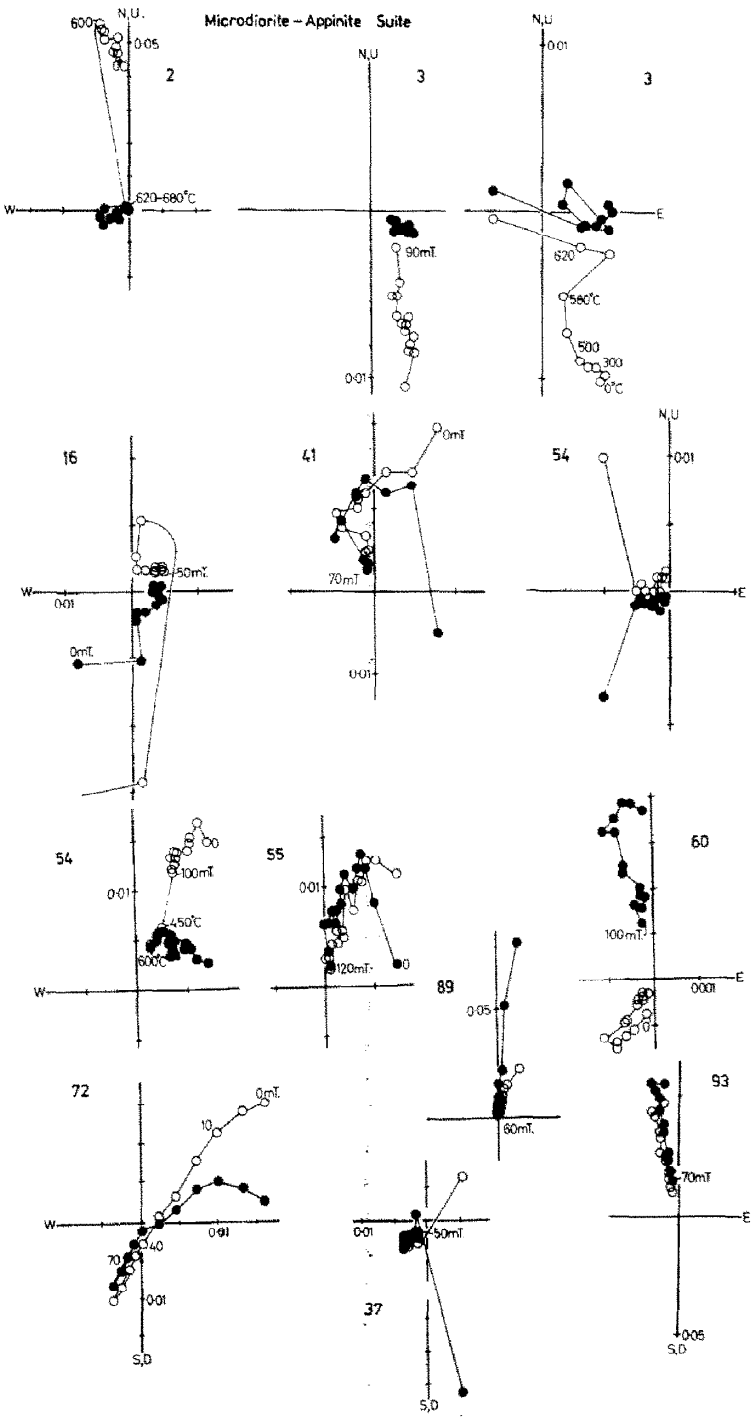



Fig. 2. Behaviour of some characteristic magnetisations within the microdiorite–appinite suite of minor intrusions with progressive a.f. and thermal demagnetisation illustrated as vector plots (Wilson, 1961; Zijderveld, 1967). Open symbols are north versus east projections and closed symbols are up versus east components.

negative directions with a.f. and thermal cleaning and comparable vectors are recognised at sites 42, 47 and 77. At site 3 the higher NRM samples move from northwest negative to southerly positive directions to group with the remaining samples which are of opposite polarity to most of the other sites in this group; site 42 (Fig. 2) illustrates a component subtracted over a broad range of a.f. treatment and unblocked above 500°C by thermal treatment. Site 2 in this figure is an example of a northerly negative component with narrow blocking temperature range unblocked near the Curie point of magnetite, and 55 is a steeper negative direction progressively recovered by a.f. cleaning. In group four sites have strongly magnetised components which diminish rapidly with treatment and 89 in Fig. 2 illustrates a typical example showing the northerly negative direction. Single components are recognised at most sites but complications include a coherent component (i) at site 53 removed at intermediate fields to recover direction (ii), and a discrete hematite-held component at site 62. Westerly negative directions at sites 56 and 58 and a possible antiparallel component at site 62 are unblocked near the Curie point of magnetite. The high coercivity group (v) exhibits good stability against treatment and although hematite is only rarely detected by thermomagnetic tests, demagnetisation behaviour and thermal treatment suggests that a hemo-ilmenite phase is an important remanence carrier here. At sites 34, 54 (Fig. 2), 93 and 110 this is the northerly negative remanence recovered elsewhere while (i) at site 54 is a lower blocking temperature component present in three different cores (see also Fig. 2). At sites 68 and 69 however, a westerly component is unblocked at about 600°C with a second higher unblocking temperature component of similar direction in three samples of 68. Remanences, mostly unblocked by the Curie point of magnetite, from several excluded sites are smeared (mostly steep southerly positive to northerly positive) in a distribution which is not effectively reduced by cleaning and may be due to incomplete removal of secondary components (the second stage thermal treatment renders this unlikely) or acquisition of the magnetisation during part of a field transition.

*Dolerites*

Thirty three sites were sampled in dolerite dykes and sheets; some are visibly metamorphosed and assigned to the amphibolite-epidiorite suite while others have no visible metamorphic characteristics. Metadolerites are mostly characterised by weak remanences apparently because most of their Fe content is incorporated in amphibole and biotite; their moments fall below levels of accurate measurement at 15–30 mT stages of treatment and three sites have no meaningful groupings. Sites 81–88 are typical of the unmetamorphosed dolerites and moderate to strong remanences exhibit moderate coercivities and coherent northerly negative directions beyond initial stages of a.f. treatment (e.g. 83 and 85 in Fig. 3); these are unblocked close to the Curie point of magnetite. Site 61 is an example of a dolerite with

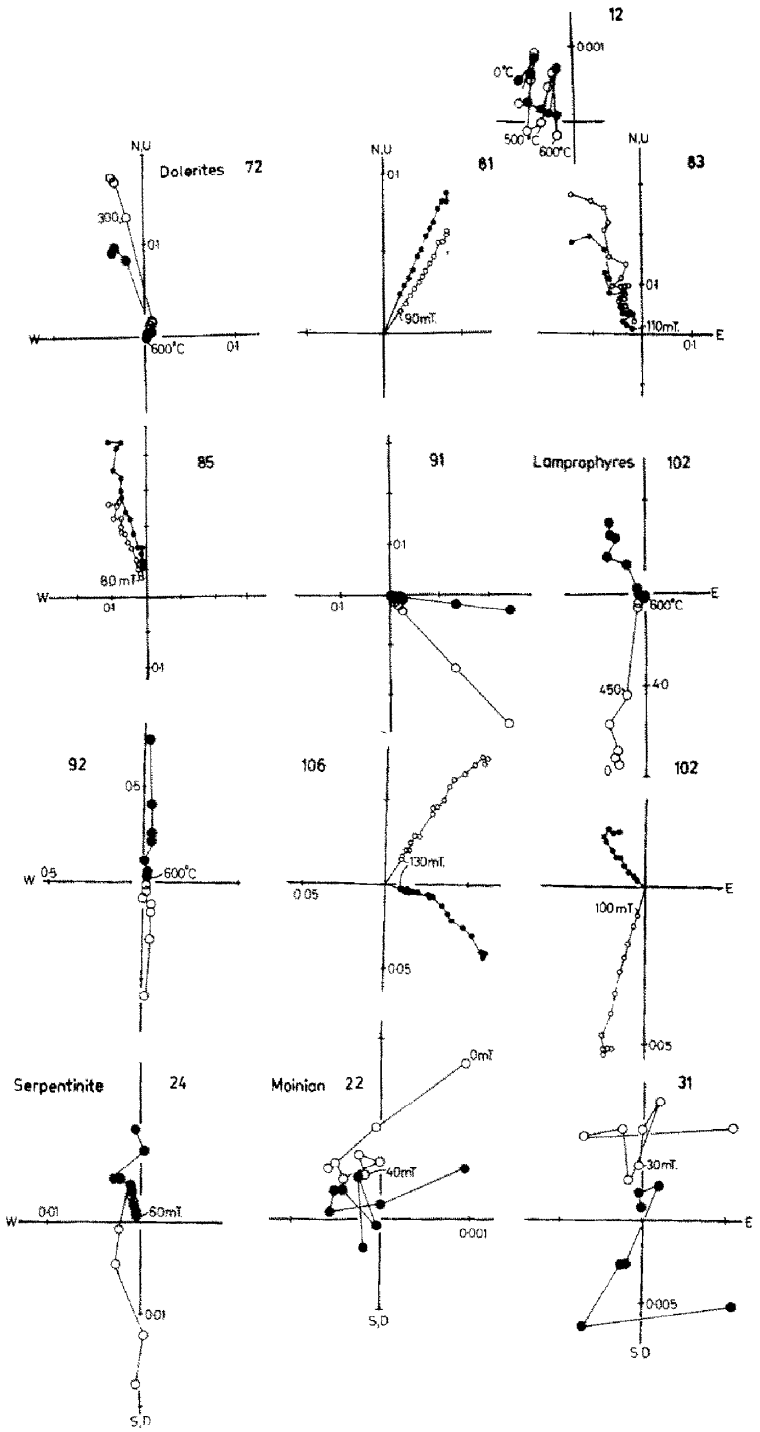



Fig. 3. Behaviour of characteristic magnetisations within some dolerite and other miscellaneous rock types with progressive a.f. and thermal demagnetisation illustrated as vector plots. Symbols are as for Fig. 2.

comparable behaviour but yielding an antiparallel direction; dual polarity is recognised at site 36. The contrasting near E–W declinations which are observed in a minority of the microdiorite suite are also present in a small number of dolerites. Site 86 is an example where a westerly shallow direction is observed as a single component in all but two cores. At site 97 it is only recovered above the Curie point of magnetite, at site 12 it is recovered by second stage thermal cleaning above the Curie point of magnetite and in one core over the whole range of treatment (Fig. 3). At site 64 directions are smeared and this site probably does not belong to this population, but at 91 a single antiparallel component is present (Fig. 3) and the same direction is recognised in three cores at site 75 following VRM removal.

*Moinian metasediments and rocks in Lewisian inliers*

Initial collections of large numbers of cores at 22 sites in Moinian sediments proved with few exceptions to be very weakly magnetised ($< 10^{-6}$ A m$^2$ kg$^{-1}$) and this aspect of the study was not pursued. However, a significant fraction of the samples in some of these sites do possess measurable and stable components (Table I). Within these large sites (which come almost entirely from thermochron zone 4 (410–400 Ma)) the coherent magnetisations are seldom entirely single but the most prominent feature is a WNW shallow remanence direction. At site 22 (Fig. 3) all cores have this as the only component present, and it is present to a lesser extent at sites 7, 21, 28, 30 and 31; in the last four cases it is not the only direction present with a component apparently belonging to the N–S axis present either in different cores (21) or recovered in some other cores at higher stages of treatment. Figure 3 shows removal of the westerly component in low fields to recover a northerly negative component. The antiparallel (ESE) direction is present as a single component at sites 20, 27 and 29; thus these Moinian metasediments contrast with the dolerite and microdiorite suites in that WNW–ESE directions comprise the majority recorded. At 10 a dolerite sheet in Lewisian yields a coherent southerly positive component unaffected by a.f. treatment and preserved to 680°C with thermal treatment. Site 4 comprises samples collected from three different rock units in a Lewisian inlier, namely an ultrabasic pod, a diorite intrusion and a late-tectonic dolerite dyke: high coercivity remanence in the first two units yielded stable NW negative directions unblocked at the Curie points of hematite in the diorite and magnetite in the ultrabasic pods; lower coercivity remanence in the dolerites unblocked by 560°C yielded scattered directions and no group mean.

*Lamprophyres*

Although the collection includes both high coercivity, high stability (e.g. 102 and 106, Fig. 3) and low coercivity, low stability (e.g. 101, 107) material, with a few exceptions the lamprophyre dykes and sheets illustrate a simple behaviour with

cleaning treatment. Samples show the progressive subtraction of one component unblocked at the Curie point of magnetite (102, Fig. 3). Since these rocks are genetically linked to the younger granites and probably post-date the regional cooling, they may have a younger magnetisation age than the remainder of the collection. However, a NW negative (58, 101) and two SE positive (105, 107) directions are defined in this division. The remaining sites (four reversed, two normal) define an axis of magnetisation $D = 193°$, $I = -19°$ ($\alpha_{95} = 13°$) equivalent to a palaeopole at 338°E, 41°S. This pole is probably of Lower to Middle Devonian age because the rock units are younger than the other intrusive suites and the characteristic magnetisation has a southerly shallow negative direction compared with predominant intermediate positive directions in the other groups. The dispersal of directions is greater however than would be expected from a single coherent sample, and it is likely that intrusion spans a considerable period of time. The lamprophyre dykes are cut on the southern sector of the Northern Highlands by east–west striking dolerite and camptonite dykes of late Lower–early Middle Carboniferous age (J. Mitchell, pers. commun., 1983) defining a coherent reversed axis of magnetisation $D = 180°$, $I = -4°$ ($\alpha_{95} = 4°$) equivalent to a palaeopole at 355°E, 35°S (authors, in prep.).

*Analysis of the Caledonian magnetisations*

The remaining data can be considered in the context of regional uplift and cooling on the assumptions that (i) the magnetisations were acquired during the uplift-related cooling and (ii) that the magnetisation has followed the sequence of K–Ar thermochrons, although the actual ages of magnetisation are undoubtedly older. This follows the procedure of McWilliams and Dunlop (1978) in the Proterozoic Grenville Province of the Laurentian Shield where the simple cooling assumption is in part confirmed by the sequence of high to low blocking temperature remanence, the remagnetisation of dyke swarms, and data from elsewhere in the shield. However, Roy and Blyth-Robertson (1979) also note some major exceptions which do not fit the simplified model and draw attention to the difficulties in assigning high blocking temperature components to the a.p.w. path because hematite may have a primary igneous or secondary metamorphic origin. Because dual polarities may not easily be recognised vector subtraction is of limited application, and further limitations of rock exposure and mineralogy control the range of mean directions found in any single study such that they frequently embrace the regional variation determined over a wider area.

In the context of this cautionary view, only a tentative application of the chrontour principle is applicable in the Caledonides as is further shown by the Grampian data (see below). Considered overall, the site mean data from thermochron zones 1–5 exhibit appreciable scatter (Fig. 4) apparently representing a true record of magnetisation over a protracted period of time because the vector plots

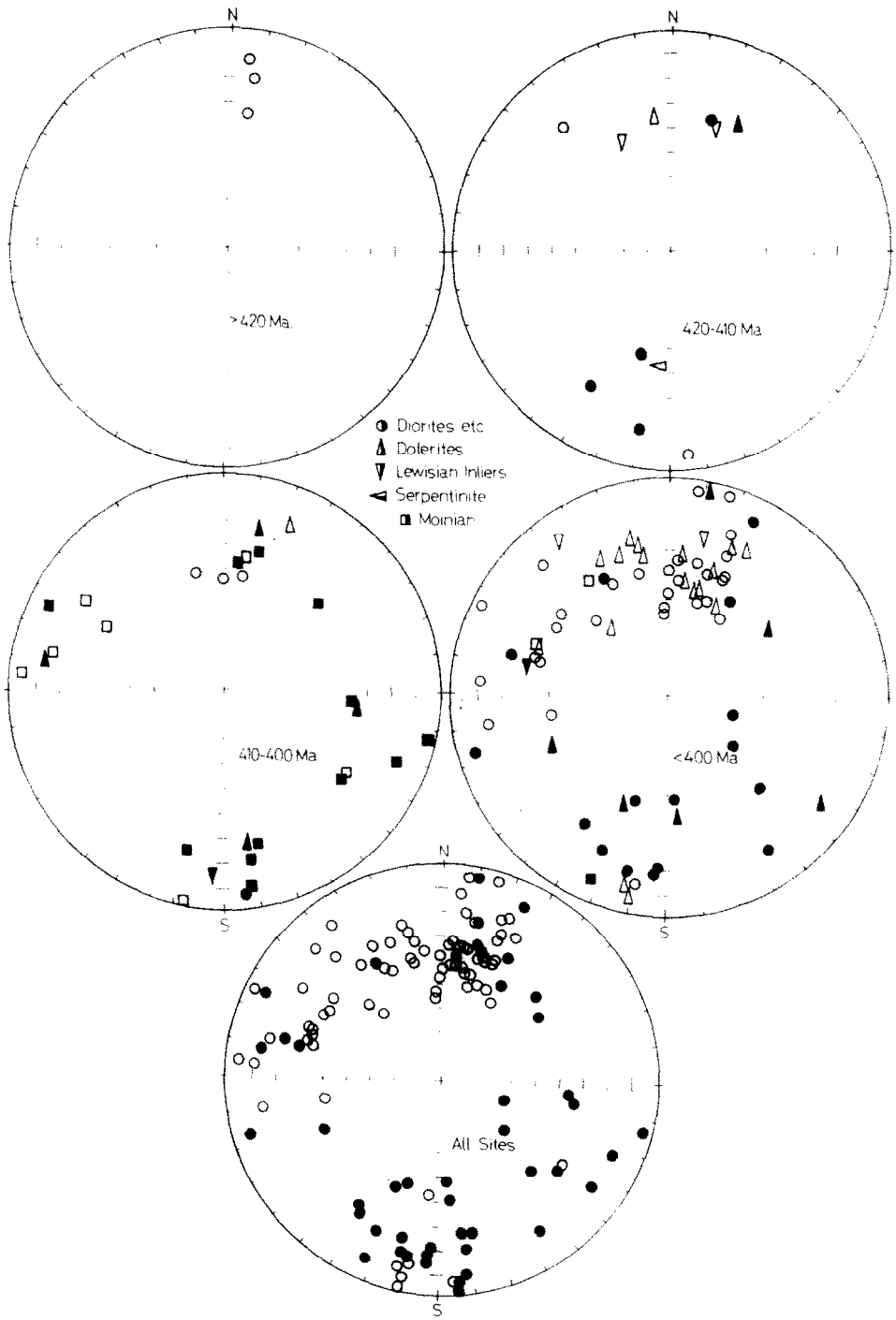



Fig. 4. Equal area plots of site group mean directions after a.f. and thermal cleaning plotted on conventional equal area plots according to the thermochron zones (Fig. 1) within which they fall; individual hematite components recognised in minor numbers of samples are excluded. Open symbols are upper hemisphere plots and closed symbols are lower hemisphere plots. The lower plot shows all site mean directions plotted as circles.

TABLE II

Group mean palaeomagnetic data for Caledonian rocks of the Northern Highlands of Scotland *

| Rock group | Sites included in calculation | $N$ | $R$ | $D$ | $I$ | $k$ | $\alpha_{95}$ | South palaeomagnetic pole | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | °E | °N | ($dp/dm$) |
| Micro-diorite-appinite suite | 1, 2, 3, 6, 8, 15, 17, 19, 25, 26, 34, 35, 40, 41, 42, 43, 44, 45, 47, 48, 50, 52, 53(ii), 54(ii), 57A, 58A, 59, 61, 63, 66, 70, 72, 77, 89, 90, 93, 94A, 100, 105, 109, 110(ii) | 41 (26N, 15R) | 38.12 | 8.6 | −29.2 | 13.9 | 6.2 | 166.4 | 16.5 | (3.8/6.9) |
| Dolerites | 10, 18, 36, 38, 39, 67(i), 70, 71, 73, 75, 78A, 79, 80, 81, 83, 84, 85, 87, 88, 94A, 97(i), 98 | 22 (18N, 4R) | 19.99 | 7.6 | −32.3 | 10.4 | 10.1 | 167.5 | 14.7 | (6.4/11.4) |
| WNW–ESE directions | 4, 5, 6(i), 7, 12, 16(ii), 20, 21(i), 22, 27(i), 28(i), 29(i), 30(i), 31(ii), 32, 33(ii), 38, 44, 46, 49, 51, 54(i), 56, 57, 58, 60, 62, 67, 68, 69, 74(ii), 75(ii), 76(i), 78A, 80A, 86, 91, 96, 104 | 39 (28N, 11R) | 35.83 | 297.8 | −21.3 | 7.7 | 8.6 | 235.6 | 4.9 | (4.8/9.1) |
| Zone 2 (430–420 Ma) | see Table I | 3 (3N) | 2.96 | 7.3 | −21.8 | 32.7 | 17.2 | 167.3 | 20.9 | (9.6/18.1) |
| Zone 3 (420–410 Ma) | see Table I | 10 (4N, 6R) | 8.50 | 10.7 | −14.4 | 6.4 | 20.7 | 163.4 | 24.6 | (10.8/21.2) |
| Zone 4 (410–400 Ma) | see Table I | 19 (10N, 9R) | 17.44 | 4.3 | −20.3 | 11.6 | 10.3 | 170.5 | 21.9 | (5.7/10.8) |
| Zone 5 ( < 400 Ma) | see Table I | 51 (33N, 18R) | 47.4 | 188.4 | 29.6 | 13.8 | 5.6 | 167.6 | 16.3 | (3.4/6.2) |
| Lamprophyres | 92, 100, 102, 103, 106, 108 | 6 (2N, 4R) | 5.73 | 192.5 | −20.9 | 18.4 | 16.1 | 338.3 | −42.4 | (8.9/16.9) |

* $D$ and $I$ are the mean declination and inclination of $N$ sites with resultant vector $R$ and 95 percent confidence circle of $\alpha_{95}$. $Dp$ and $dm$ are the semi axes of the 95 percent confidence oval about the palaeomagnetic pole in the colatitude direction and at right angles to it respectively.

(e.g. Figs. 2 and 3) show that single components are being subtracted by cleaning treatment at most sites. Since the maximum temperature reached during late tectonism is close to the Curie points of the high blocking temperature phases (Richardson 1968), at least some of these units may have a primary igneous magnetisation and a long time averaging effect cannot be assumed here.

The overall distribution shows a predominant grouping of NNE negative directions, with a sparse anticlockwise negative distribution around to the subordinate WNW directions. This distribution is complemented by antiparallel positive directions ranging from the predominant SSW direction to the subordinate ESE directions. The palaeopole migration equivalent to the groupings of directions identified in these rocks must accommodate the NNE and WNW dual polarity axes. The path between them is represented by relatively few data points and was probably executed rapidly. The calculation of representative palaeopoles is inevitably subjective but because the number of data points is relatively large the criteria only marginally effect the position of the group mean. We adopt a general basis for palaeopole calculations without rigid exclusion criteria: the only directions excluded are firstly hematite components present in small numbers of samples at some sites which mostly have southerly intermediate negative remanences (e.g. 64) conforming with probable Middle Devonian to Permian palaeofield directions and apparently of secondary origin. Secondly we exclude a number of northerly positive components (e.g. 23) which do not survive advanced a.f. or thermal treatment and probably include a Mesozoic to Recent component. In Table II the predominant direction from the microdiorite suite accommodates all directions (or their antiparallel equivalents) between $D = 316°$ and 34°E, and $I = +14°$ and $-57°$; the predominant direction from the dolerites accommodates all directions falling between $D = 333°$ and 56° and $I = +7°$ and $-50°$. The subordinate NNW–SSE axis is defined by all directions falling between $D = 253°$ and 331° and $I = +32$ and $-61°$; positive components at sites 17 and 18 are excluded because they are eliminated by low field a.f. treatment. The palaeopole estimates are thus general rather than specific. They may obscure some facets of the magnetisation history, but we believe that they provide a more realistic basis for interpretation than palaeopoles calculated from small numbers of sites; this is especially the case because it is not clear whether the remanences date from primary igneous cooling or are linked to uplift related cooling, and because the extent to which the magnetisations were involved in the last stage (4) of the regional folding is also unclear. In both cases the wide areal extent of the sample helps to overcome these uncertainties surrounding the magnetisation history. To test the possible influence of regional cooling on the magnetisation history we also group sites falling in the predominant group (considering declinations between 320° and 35°E) according to their situation within the K–Ar muscovite age chrontour zonation of Dewey and Pankhurst (1970) to derive the representative mean chrontour palaeopoles of Table II.

The subordinate WNW–ESE axis of magnetisation is unlike any post-Silurian

palaeofield direction for Britain, and since directions of both polarities are represented it would appear to define a representative palaeopole for this region. It differs from the predominant direction only in terms of declination and could thus be accommodated by in situ rotation of the sample area. The relationship of this direction to the predominant field direction is not completely clear because it normally occurs as discrete single components. As previously noted these are subordinate in the igneous suites but predominant in one part of the Moinian outcrop. At sites where the directions occur associated with other components they are either in different cores from the same sites in Moinian metasediments [27(i), 28(ii), 29(i)] where the components are too weak for their origin to be reliably established, or high blocking temperature components in a smaller number of cores [sites 12, 31, 33(ii)] following the unblocking of the predominant component. Elsewhere these directions are entirely in components unblocked by the Curie point of magnetite or have properties suggesting that they are magnetite-held; the two exceptions are sites 4 and 44 where the components are apparently hematite-held. Thus the present data do not unequivocally define the age relationship between the NNE and WNW axes and the swathe between them, but from the subordinate nature of the WNW axis and its prominence in the Moinian rocks of one of the youngest chrontour zones, we provisionally suggest that it is younger than most, or all, of the magnetisations contributing to the predominant remanence.

## INTERPRETATION

Two important uncertainties surround the magnetisations carried by the rocks of this area. These are namely whether or not the magnetisation date from primary igneous cooling or uplift-related cooling, and, if the latter, by how much they pre-date the closure times indicated by the isotopic studies. If the magnetisations are linked to igneous cooling they are at least 420 Ma and possibly as much as 450 Ma old. Given that age differences of about 10 Ma between the closure times of muscovite and biotite are typical in this terrain (Harper 1967, Dewey and Pankhurst 1970), the magnetisation ages, and accordingly the palaeomagnetic poles of Table II, are also inferred to be about 20–30 Ma older than the isotopic ages and applicable to Ordovician or Silurian times. Precise assignment to the stratigraphic column is uncertain because of doubts about the absolute age boundary between the Ordovician and Silurian periods: a recent discussion of the Lower Palaeozoic time scale by Gale et al. (1979) is disputed by McKerrow et al. (1980) who regard several key Rb–Sr ages on silicic extrusive rocks as records of metamorphic, and not igneous, events and place the Ordovician–Silurian boundary as low as 437 Ma in accord with the 435 Ma assignment of Lanphere et al. (1977) based on K–Ar determinations. However, Rundle (1981) presented more data confirming the scale of Gale et al. to place the Ordovician–Silurian boundary at ca. 420 Ma; the Silurian is regarded as a short period of only 10–15 Ma duration. This may be too young: Harland et al.

(1982) place the Ordovician–Silurian boundary between 438 and 428 Ma, an interval incorporating the stratigraphic division to which British poles 23–27 in Fig. 5B are assigned, while Compston et al. (1982) reinterpret Rundle's critical Ashgill-age isochron as 430 ± 7 Ma. It thus appears that the palaeopoles from the study are to be assigned to Upper Ordovician and/or Lower Silurian times. An assignment to Middle Silurian–Lower Devonian times is also less likely because typical remanence directions for the interval (Briden et al., 1973; Piper, 1979) are only rarely found (Fig. 4) although they are known to be present in the southwestern sector of this tectonic division (Watts 1982).

To evaluate the tectonic significance of these palaeopoles they are considered in Figs. 5A–C in the context of the contemporary palaeomagnetic data from other tectonic divisions of the Caledonian orogenic belt and the bordering cratonic forelands of North America–Greenland and Scandinavia. The data are summarised in the figure captions and subdivided according to the structural province from which they originate; they are plotted here on the conventional reconstruction of the North Atlantic continents (Bullard et al., 1965) with North American data rotated anticlockwise towards Europe by 38° about the Euler pole at 27.7°E, 88.5°N. Since the assigned primary or secondary ages of magnetisation of many of these units are open to alternative interpretations, this and other analyses should be treated cautiously and referred to the original data sources. Britain is treated as a coherent entity in palaeomagnetic terms between the GGF in the north and the Hercynian Front in the south (Briden et al., 1973) although further work may recognise this as an oversimplified assumption. In Scandinavia two structural divisions are recognised namely the Fennoscandian Shield and the zone of allochthonous units to the west of Caledonian thrust zone; few data points are currently available here and no differences are yet apparent between these two divisions. Following Williams (1978a, b) five structural divisions are recognised in the Appalachian orogen bordering in sequence the stable Laurentian block of North America and Greenland. The continental margin of eastern North America is a Palaeozoic sedimentary wedge deposited on an ancient continental margin previously subjected to rifting and magmatism in latest Precambrian times. The Iapetus suture zone to the east is the consumed relict of an ocean basin of primary Cambrian and early Ordovician age (Harland and Gayer, 1972; Dewey, 1974) linked with the obducted ophiolites in Ireland, the margins of the Midland Valley in Scotland, the Shetland Islands and the Norwegian Caledonides. The Avalon microcontinent comprising eastern New England and parts of the Maritime Provinces of Canada is a crustal block with a basement of low-grade volcanic and sedimentary rocks of Late Precambrian age, and to this extent is analogous with the basements of England and Wales; it was welded to the orogen following closure of the Iapetus Ocean and strike-slip motions of disputed magnitude during the (Devonian) Acadian orogeny (McKerrow and Ziegler, 1972). Still further to the east are segments of the Appalachian Piedmont and Nova Scotia with geological relationships not readily linked to the Avalon Zone. It is

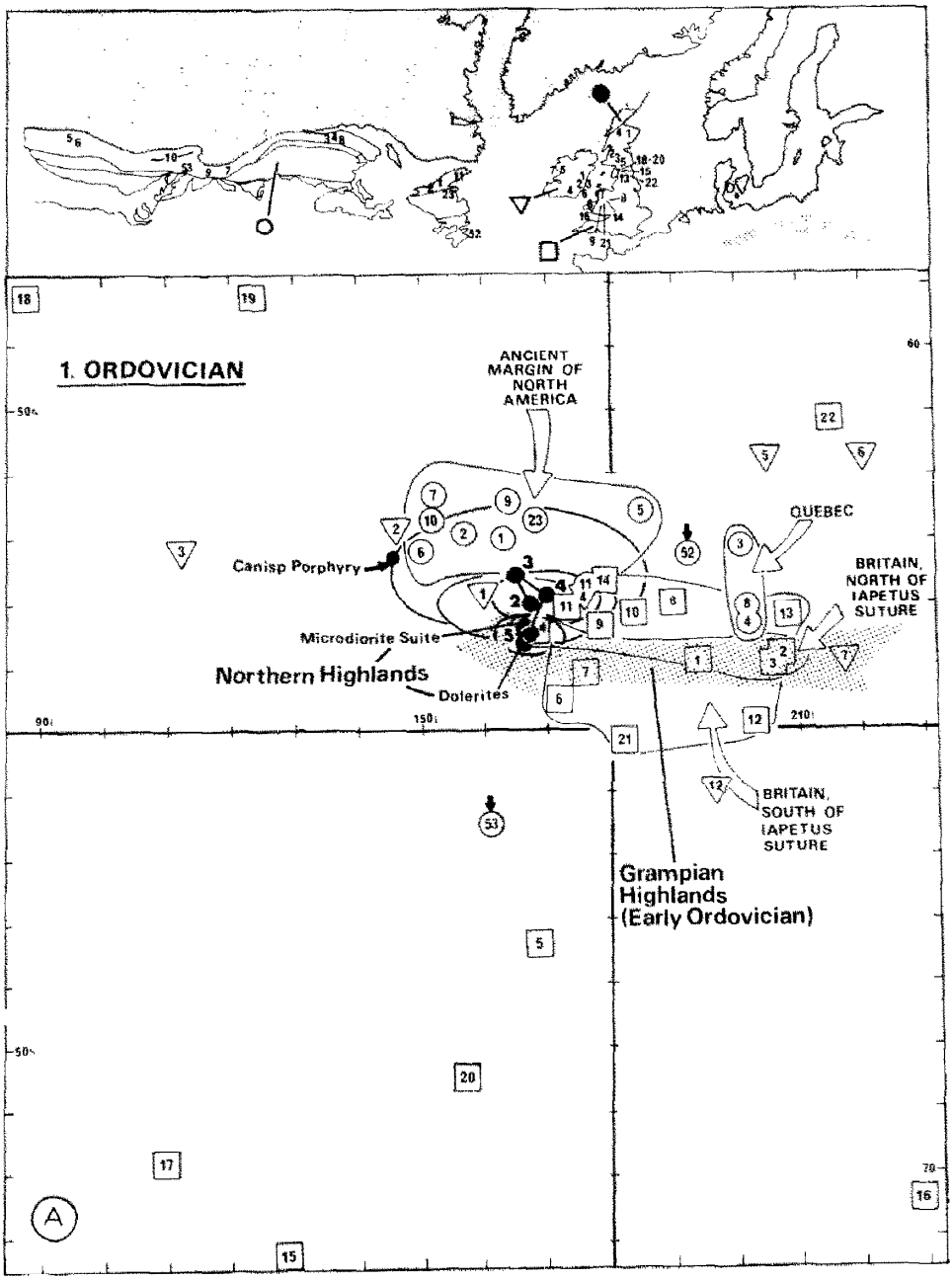
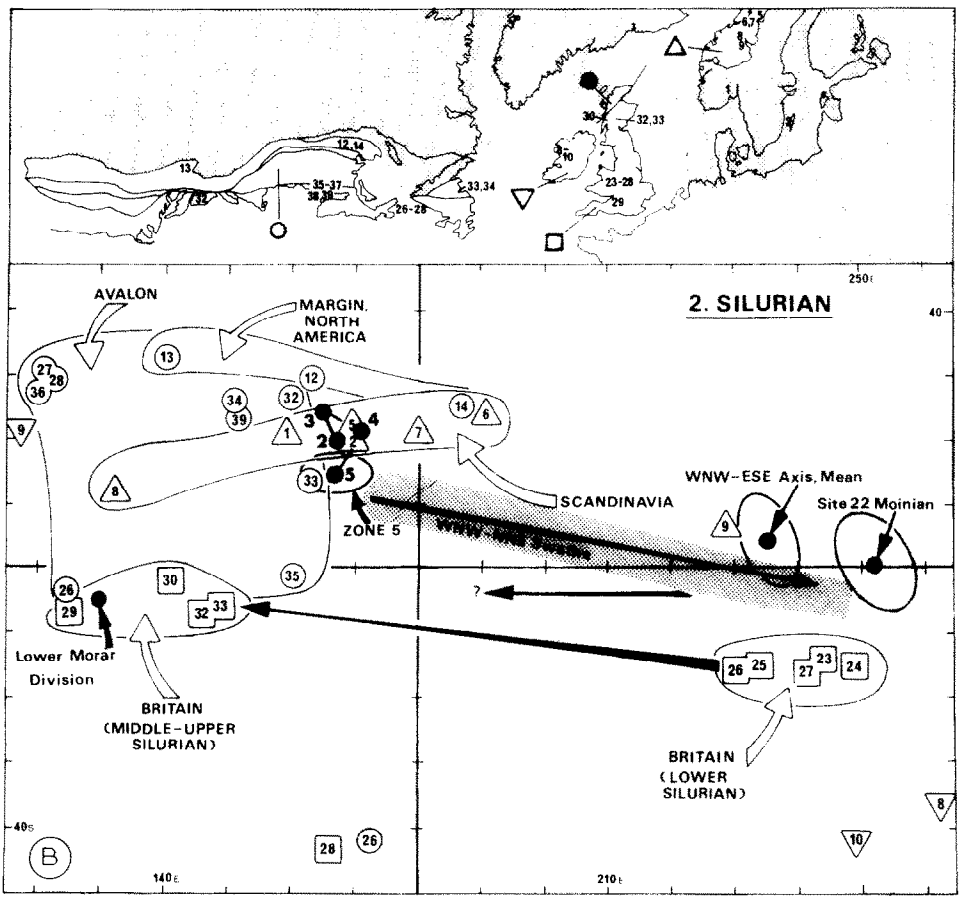



Fig. 5. Palaeomagnetic poles from the Appalachian–Caledonian orogen and adjoining areas (North America = circles, Ireland = inverted triangles, Britain = squares and Scandinavia = triangles), assigned to (A) the Ordovician, (B) Silurian and (C) Devonian periods, are plotted on the pre-drift reconstruction (Bullard et al., 1965). Group mean palaeopoles from the Northern Highlands of Scotland (solid circles) are plotted for comparison. Ages of the poles are indicated as O = Ordovician, S = Silurian, D = Devonian,




C = Carboniferous, l, m, u = lower, middle and upper divisions of these periods, ar = Arenig, ln = Llanvirn, ld = Llandeilo, c = Caradoc, a = Ashgill, lv = Llandovery, we = Wenlock.

The poles have been grouped into seven tectonic divisions (a)–(g), see text—and numbered according to the following listing:

(a) *Ancient continental margin of eastern North America* (○)

*1*—St. Georges dolostone, Newfoundland, O(l); *2*—St. Georges limestone, Newfoundland O(l); *3*—Spider Lake-Woburn igneous suites, O(l); *4*—Weedon Gould area rocks, first pole, O(l–m); *5*—Mocassin and Bays Formations, O(l–m); *6*—Chapman Ridge Formation, O(l–m); *7*—Trenton Limestone O(m); *8*—Quebec Volcanics, O(m); *8*—Beemerville Alkaline Complex O(u); *10*—Juniata Formation, O(u); *11*—Steel Mountain Volcanics, Newfoundland, O; *12*—Weedon-Gould rocks, second pole, S(m?); *13*—Bloomsburg Formation (S); *14*—Weedon-Gould rocks, third pole, S–D(?); *15, 16*—Mount Peyton rocks (420 and 380 m.y.); *17*—Mt. Aylmer rocks, D(m–u); *18*—Scotstown intrusion D(m–u); *19*—Winslow granite, D(m–u); *20*—Winslow hornfels, D(m–u); *21*—Megantic area rocks, D(m–u); *22*—Spider Lake-Woburn area rocks, D(u).

(b) *Iapetus suture zone, Andean margin of eastern North America and the Avalon microcontinent* (○)

23—Moreton's Harbour Volcanics, Newfoundland, O(l–m); *24, 25*—Bonavista Bay alkali dykes, Newfoundland, 400–344 m.y.; *26, 27, 28*—Cape Breton Granites, S–D, 450–350 m.y.; *29, 30, 31*—St. Cecile–St. Sebastian rocks, D(m); *32*—low-grade metavolcanics, Massachusetts, 420–400 Ma; *33, 34*—Botwood Group, S(l); *35, 36, 37*—Mascarene intrusives, greywackes and siltstones, S, 420–400 Ma;

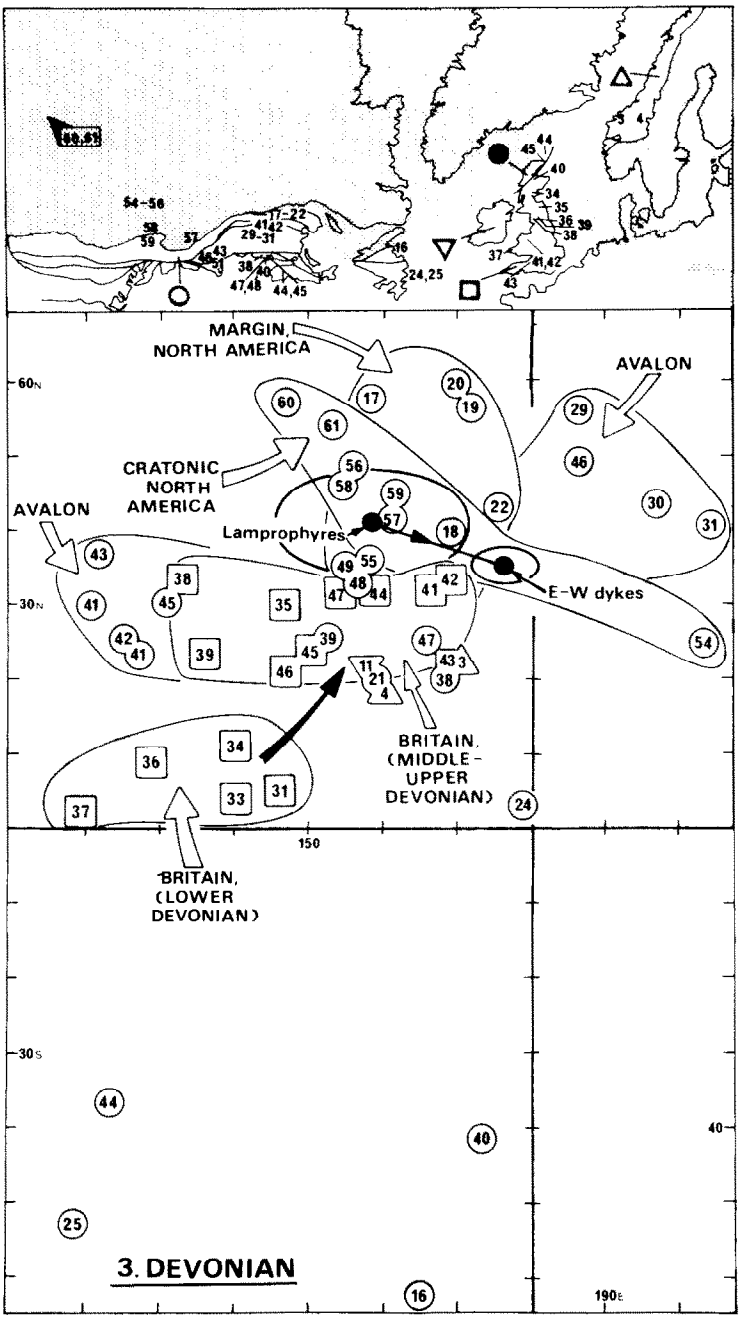



Fig. 5C

*38*—Eastport Formation, D(l), 412 Ma; *39*—Hersey Formation, S(u)–D(l); *40*—St. Stephen Pluton, 408 Ma; *41*—Traveller felsite, D(1); *42*—Dockendorff Group, D(l); *43*—Hartwick diorite, 400 Ma; *44, 45*—St. George pluton, 394 Ma; *46*—Belcherton intrusion, 380 Ma; *47*—Perry lavas, D(u); *48*—Perry sediments, D(u); *49*—Maringoan Formation, C(l); *50*—Hopewell Group, C(l); *51*—Mafic complex, Massachusetts, 370 Ma.

Fig. 5 (*continued*).

(c) *Possible ancient NW African margin of Gondwanaland* (○)

52—Wabana Group, Newfoundland, O(l); *53*—Arden pluton, 440 Ma.

(d) *Reference poles from cratonic North America* (○)

*54*—Columbus Limestone–Raisin River dolomite, D(m); *55*—Columbus limestone, D(m); *56*—Delaware limestone, D(m); *57, 58, 59*—Catskill Formation, three studies, D(u); *60*—Becker Butte rocks, D; *61*—Temple Butte rocks, D.

(e) *Caledonides of Ireland* (▽)

*1*—Grangegeeth Volcanics, O; *2*—Portrane Volcanics, O; *3*—Balbriggan and Sherricks Volcanics, O; *4*—Kildare inlier volcanic rocks, O; *5*—Glensaul felsite, O; *6*—Lambay Island andesites, O(u); *7*—Mweelrea ignimbrites, O(l); *8*—Salrock Group, S(u); *9*—Knocknaveen Group S(u); *10*—Intrusives and baked contacts, Killary Harbour, S(u); *11*—Derry Bay felsite, D(u); *12*—Tramore Volcanics, O(m).

(f) *British Caledonides* (□)

(i) *Ordovician rocks from the northern margin of the Iapetus suture.* *1*—Younger gabbros of Aberdeenshire, 486–461 Ma, (Rb–Sr), 460–420 Ma (K–Ar); *2*—Ballantrae Complex, serpentinites, 475 Ma, O(ln–ld); *3*—Byne Hill gabbro, O(ln–ld); *4*—Foyers Complex, 460–420 Ma; *5*—Bail Hill Volcanics, O(c).

(ii) *Ordovician rocks from the south of the Iapetus suture.* *6*—Trefgarn Andesitic Series, O(ar); *7*—Eycott Group, O(ln); *8*—Stapeley Hill Group, O(ln); *9*—Builth Volcanic Series, O(ln); *10*—Carrock Fell Complex, 468–439 Ma, O(c); *11*—Round Knott dolerite, O(c); *12, 13*—Borrowdale Volcanic Group, lower and higher horizons, 468 m.y., O(ln–ld); *14*—Tuffs, Welsh Borderlands, 451 Ma O(c); *15*—Stockdale rhyolite, 430 Ma, O(a); *16*—Cader Idris basalts, O(c?); *17*—North Wales igneous province combined result; *18*—Threlkeld–St. John's microgranodiorite, 439 Ma, O(ln?); *19*—WNW dolerite dykes, North Lake District; *20*—WNW Kersanite dykes, Cross Fell, O–S ( < ln); *21*—dolerite intrusions Builth Inlier, O(c–a); *22*—Cautley Fell rhyolite, O(a).

(iii) *Silurian rocks from Britain (all south of Great Glen Fault zone).* *23*—Andesite intrusions, Shelve inlier, O–S (l–lv); *24*—Corndon Hill phaccolith, O–S (a–lv); *25*—dolerite intrusions, Shelve Inlier, O–S (a–lv); *26*—E–W dolerite dykes Welsh Borderlands, S(lv); *27*—Moel-y-Golfa andesite S(lv); *28*—Breidden Hill dolerite, O–S(a–lv); *29*—Somerset and Gloucestershire lavas, S(we); *30*—Lorne Plateau lavas, 421 Ma, S; *31*—Lower O.R.S. lavas Glencoe, S–D; *32*—Arrochar Complex, 418 Ma; *33*—Garabal Hill—Glen Fyne intrusions, 404 Ma, S–D.

(iv) *Devonian and Carboniferous rocks (from both sides of the Great Glen Fault Zone).* *34*—Lower O.R.S. lavas and sediments, Midland Valley of Scotland, S–D, 394 Ma; *35*—Lower O.R.S. lavas, Cheviot, 390 Ma, S–D; *36*—Cautley Fell dolerite; *37*—Fishguard Volcanic Group secondary magnetisation; *38*—Shap granite, 393 Ma; *39*—SW–NE minette dykes, Cross Fell; *40*—Middle O.R.S. sediments Foyers; *41*—Hendre dolerite, D; *42*—Blodwell keratophyre, D; *43*—Upper O.R.S., Bristol district, D–C; *44*—Orkney lavas 383 Ma, D(m); *45*—Caithness Sandstone, D(m); *46*—Skomer Volcanic Group (Hercynian secondary magnetisation), D(m)–C.

(g) *Poles from Scandinavia* (△)

(i) *Reference poles from the Fennoscandian platform.* *1*—Ringerike Sandstone, S(u); *2*—Gotland limestones, S(u).

(ii) *Poles from the Caledonides:* *3*—Kvamshesten sediments, D(m–u); *4*—Roragen sediments, D; *5*—Sarv nappe dolerites, Caledonian remagnetisation, S–D; *6*—Mosjoen gabbro 'A'; *7*—Sulitjelma gabbro 'A', S(u)–D(l); *8, 9*—Fongen-Hyllingen gabbro P1 and P2 components, D(1), 405–400 Ma.

The North American poles are compiled by Kent and Opdyke (1978), Van der Voo and French (1977), Roy et al. (1979) and Rao et al. (1981). The Irish poles are taken from Morris (1976) and listings of the "Geophysical Journal of the Royal Astronomical Society". The British poles are compiled by Briden et al. (1973) and Piper (1979). The Scandinavian poles are summarised by Bylund and Zellman (1981); poles *6* and *7* are unpublished data of authors.

In Fig. 5A the stippled arrow illustrates the a.p.w. path identified from uplift magnetisations of early

speculated (Rankin 1975) that they represent fragments of the northwest African margin of Gondwanaland accreted during the Acadian orogeny and transferred to the Laurentian block following Mesozoic rifting; the two poles from this zone are indicated by arrows in Fig. 5A; only one is aberrant.

With the exception of poles 15–20, the data from Britain comprise a coherent group of palaeopoles centred at 190°E, 15°N and mostly of Llanvirn and Llandeilo age (ca. 470–435 Ma). Pole *1* from the Younger Gabbros of Aberdeenshire is now regarded as a time-average of the palaeofield more precisely defined by an east to west swathe (Watts and Briden 1981). The palaeopoles *15–20* widely discordant from the remaining data are probably of late Ordovician (Caradoc or Ashgill) age; a tectonic explanation appears to be inapplicable and they have been provisionally considered in the context of either rapid a.p.w. movements or non-dipole behaviour (Thomas and Briden, 1976; Piper, 1979).

Lower to Upper Ordovician palaeopoles from platform sediments along the ancient continental margin of North America (poles *5–7*, *10*) comprise a grouping distinct from the British poles and centred at about 160°E, 30°N (Fig. 5A); the case for the primary nature of some of these units has been extensively argued (McElhinny and Opdyke, 1973; Van der Voo and French, 1977) although the age of magnetisation prior to Silurian folding is not normally constrained by field tests. The Lower to Middle Ordovician units from the margin of the orogen 1500 km to the northeast in Quebec yield aberrant poles *3*, *4* and *8*, but a tectonic explanation is probably inapplicable here because poles *1*, *2* and *11* from the continuation of this zone in Newfoundland accord with the Appalachian group.

British Silurian palaeopoles fall into two groups (Fig. 5B): the first (22–26) is of late Ashgill to late Llandovery age and derived from a small sector of the paratectonic Caledonides in Wales (Piper 1978); this is regarded as part of the overall a.p.w. path because there is no evidence that the area is allochthonous and because older Ordovician palaeofield directions here are comparable with those from elsewhere. Post-Llandovery palaeopoles lie near 140°E, 5°S and the latitudinal difference of 15°–25° between the British and North American poles persists in the Silurian data. Poles *1* and *2* from the Fennoscandian platform conform with poles *5–7* from the Scandinavian Caledonides and fall in the same field as the North American palaeopoles to confirm that the two regions had approximately welded into the Pangaean configuration by Upper Silurian times (McKerrow and Ziegler, 1972). The two remanence directions identified in the Fongen-Hyllingen gabbro in the Scandinavian Caledonides (Abrahamsen et al., 1979) may record magnetisation

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Ordovician age in the Grampian Highlands on the south side of the Great Glen Fault (Watts and Briden, 1981; Sallomy and Piper, 1973a); this path (see text) is interpreted to predate the path identified in this work from the Northern Highlands. In Fig. 5B the stippled swathe is the path represented by the distribution to the WNW–ESE magnetisations identified in the Northern Highlands and probably of Late Ordovician early Silurian age.

during a local tectonic rotation or gross a.p.w. motion as defined by the British data (Fig. 5B).

The Lower Devonian palaeopoles from Britain (*32–36*) plot in a field distinct from poles of known or probable Middle and Upper Devonian age (*39–46*) and only marginally closer to them than the Middle–Upper Silurian poles (*28–31*). The Middle–Upper Devonian poles, however, overlap with the contemporaneous North American and Scandinavian data. It is this latitudinal discrepancy which, if it is to be explained in tectonic terms, requires a short term relative movement in Lower–Middle Devonian times equivalent to a sinistral strike-slip motion of 2500–3500 km (Morris 1976, Piper 1979).

The palaeopoles from the Northern Highlands of Scotland plot precisely between the distributions of British and North American Ordovician palaeopoles (Fig. 5A) and are not significantly different from either. A sinistral movement along the GGF cannot now be invoked to explain the differences between these two groups: the tectonic grounds for this are definitive because there are no major dislocations between the northern Highlands and cratonic North America on the conventional reconstruction which could accommodate large relative movements other than the Moine Thrust Zone; uncertainty in the position of the Lewisian forelands is limited to a motion of less than about 200 km to the southwest by the constraint that it was not reworked during the Grenville mobile episode at ca. 1100 Ma (Bridgwater et al., 1973). The absence of any significant difference between these results from the Scottish Northern Highlands and the North American data is further shown by data from the Canisp Porphyry, part of a silicic suite intruding Torridonian and Cambrian rocks on the foreland with related rocks also seen to cut the Lower Ordovician (Arenig) Durness Limestone. The pole position from 16 sites is at 146°E, 28°N (result of Dr. C. Thomas, listed by Morris, 1973); it plots close to the Ordovician data points from North America although error limits are large: ($dp/dm = 11/22$).

Given this clear compatability, we are left to question why there should be a significant difference between the remainder of the British and North American Ordovician palaeopoles in Fig. 5A. The British data comprise at least three different components: firstly there are the data points from the paratectonic Caledonides which are rotated with respect to the North American data by up to 30° anticlockwise and signify a post-Caradoc rotation of the British subplate by this approximate amount; the precise extent of this rotation cannot yet be defined because the new data from Quebec show (as the British data has done already) that appreciable a.p.w. is present in the Ordovician interval.

Secondly, part of the latitudinal difference in the poles plotted in Fig. 5A can be accommodated by Ordovician closure across the Iapetus suture. The Ilanvirnian Tramore volcanics (*12*) from the southern margin of the suture in Ireland differ from the contemporary Mweelrae volcanics (*7*) on the northern margin to an extent which has been interpreted by Deutsch (1980) in terms of a wide Iapetus Ocean (~ 3000 km). The estimate originally made from the British data (Briden et al. 1973)

is no longer valid because the British poles from the north (*1–3*) and south (*6–11*) sides of the suture zone are of Lower and Middle Ordovician ages respectively. Thus the slight latitudinal differences between some of the British Ordovician poles and the North American–Scottish Northern Highlands data in Fig. 5A could include a component of post-Middle Ordovician closure across the Iapetus suture.

Thirdly, the palaeomagnetic data from the Aberdeenshire Gabbros in the Grampian Highlands south of the GGF (Sallomy and Piper, 1973a) have been reinterpreted in the context of uplift-cooling magnetisation processes linked to K–Ar chrontour ages of 468–430 Ma (Watts and Briden, 1981). The resultant a.p.w. path represents a clockwise movement of the palaeofield vector from approximately $D = 140°$ to 190° and inclination increase from $I = 25°$ to 40°. This path cannot be directly compared with the path from the Northern Highlands because it is represented entirely by remanence of reverse polarity. Indeed, it is likely to represent a time interval shortly following early Arenig deformation (ca. 475–465 Ma) in the Highlands (Dewey and Pankhurst, 1970) both because palaeomagnetic studies elsewhere show that an appreciable part of the early Llanvirn to late Llandeilo interval was normally magnetised (Briden et al., 1973), and because the Grampian data incorporate the reversely magnetised Ballantrae ophiolite complex data points (*2* and *3*) dated ca. 475 Ma. This observation casts considerable doubt on the validity of applying a thermochron analysis to the orthotectonic Caledonides and implies a preference for the suite means listed in Table II. The resultant path from the Grampians has the same latitudinal position as, and terminates close to, the path derived here from the Northern Highlands and estimated to have a maximum age of about 450 Ma (Fig. 5A).

Comparison between the Silurian data is shown in Fig. 5B. The Northern Highlands data are significantly removed from the remaining British data points defining the two groups of Lower Silurian and Middle Upper Silurian age, although the WNW–NNE swathe reproduces, but in displaced position, the path between the British Ordovician and Lower Silurian poles. The minority WNW–ESE axis has a similar longitudinal position to, but is removed by 20° in latitude from, the British poles of Lower Silurian age; it is highly likely that these magnetisations are of comparable age (ca. 435–420 Ma). The comparative British poles are of stratigraphically-defined late Ashgill to late Llandovery age, and the same a.p.w. motion is indicated by poles from Ireland (*8*, *10*) and Scandinavia (*8*), also of probable Silurian age (Fig. 5B).

It is also clear that no significant palaeomagnetic difference between the Northern Highlands and the rest of Britain remains after post-Lower Silurian times because the Moinian rocks of the Lower Morar Division yield palaeopoles identical to other British poles falling in the Middle Silurian to Lower Devonian group (Watts, 1982; the palaeopole in Fig. 5B is calculated from dual polarity sites *1*, *2*, *3*, *4*, *5*, *6*, *8* and *9*). McClelland-Brown (1982) also recognises this remanence in the Strontian Granite. This accordance precludes major post-Lower Silurian movement along the

GGF. The difference between the Lower Silurian poles and the WNW–ESE axis *could* be interpreted in the context of about 2000 km of sinistral motion along the fault zone in Lower Silurian times. This interpretation is not sustained by the older Ordovician data (Fig. 5A) and since the Lower Silurian difference represents a rapid polar shift constrained by limited data over a short time interval, we prefer a conservative interpretation that does not require large relative movements along the GGF at this time.

The accordance of the palaeomagnetic record from the Northern Highlands with Britain south of the GGF requires that differences between the British and North American post-Ordovician data (Figs. 5B and 5C) have a non-tectonic cause. This general conclusion is further emphasised by accordance of poles from the lamprophyres and E–W dykes in the Northern Highlands with the data field for cratonic North America while they lie at the extremity of the distribution of Middle Devonian–Lower Carboniferous poles from Britain (Fig. 5C). It has the consequence that the palaeopole positions defined by British Lower Silurian and Middle Silurian–Lower Devonian data have not yet been discovered in North America. This may be either because this time period has been inadequately covered or because of the prevailing acute problem of Acadian remagnetisation in the Appalachian terrain (Roy, 1982). The consensus of the collective data is then that the Ordovician to Devonian interval was a period of appreciable a.p.w. movement, a conclusion which has already emerged from the Australian data (Goleby, 1980) where the critical a.p.w. motions Ordovician → Lower–Middle Silurian → Middle Silurian–Lower Devonian are not dissimilar from the path shown for Britain in Fig. 5B.

This study has little new information to contribute to the reason for the difference observed by Scotese and Van der Voo (1982) between Middle Devonian to early Carboniferous data from either side of the GGF. Since both geological arguments (Donovan et al., 1976; Winchester 1973) and palaeomagnetic studies (this work; Watts 1982; McClelland-Brown, 1982) exclude 2000 km of intra-Carboniferous movement, non-tectonic explanations must be sought for these differences. We favour an explanation in terms of a.p.w. movements and/or non-dipole components added to the main field: complexities of a.p.w. and field behaviour in Palaeozoic times have generally been inadequately accommodated by workers who seek to explain differences in poles in tectonic terms. Sallomy and Piper (1973b) found important minority directions of magnetisation in Devonian lavas of the Strathmore region of Scotland in addition to the main dual polarity axis, and Thomas and Briden (1976) derived an important suite of anomalous Ordovician poles from the Welsh Igneous Province; neither example satisfies a tectonic explanation.

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